

STM32 Quadcopter Flight Controller (FC) V2.0

STM32四轴飞行器飞控板系统设计

HARDWARE DESIGN PORTFOLIO

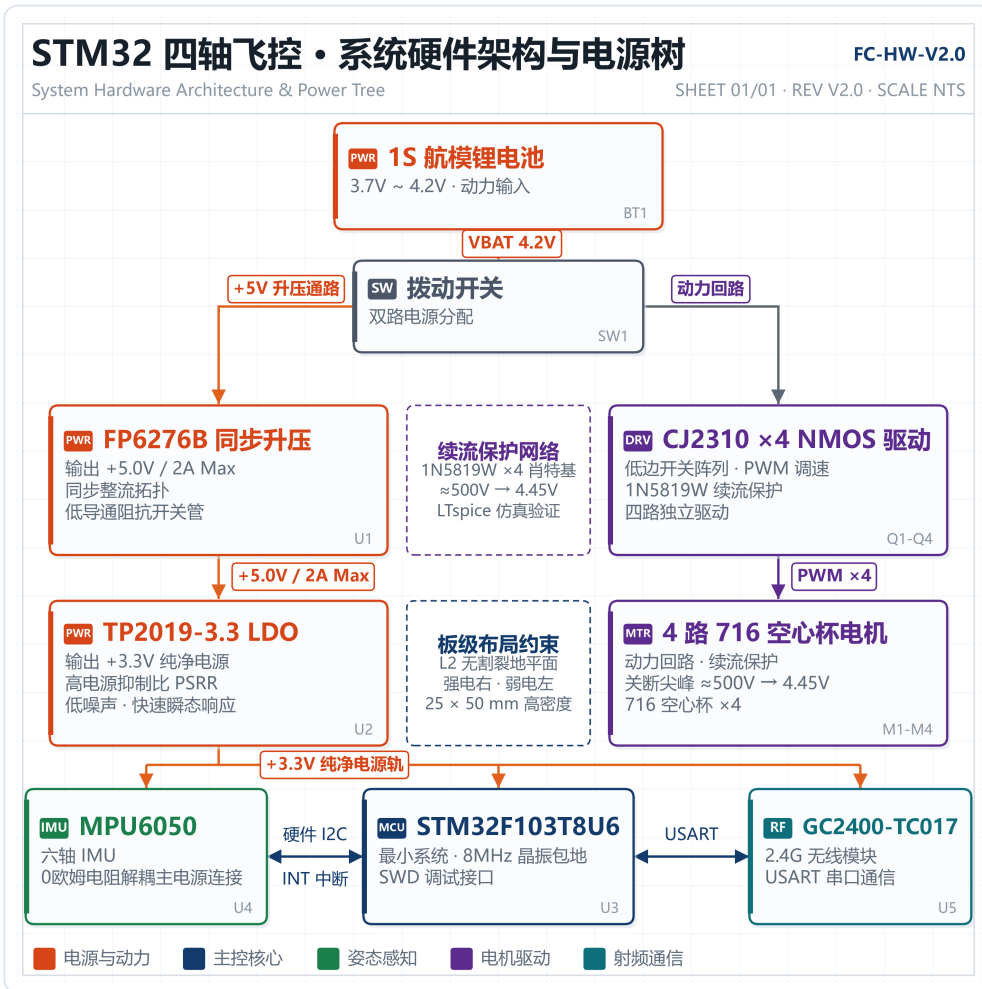
PROJECT OVERVIEW · SHEET 01 / 13

4-Layer PCB

4.45V Simulation Result

Hardware Design Portfolio

系统硬件架构与电源树 / SYSTEM ARCHITECTURE & POWER TREE



核心参数 / KEY SPECIFICATIONS

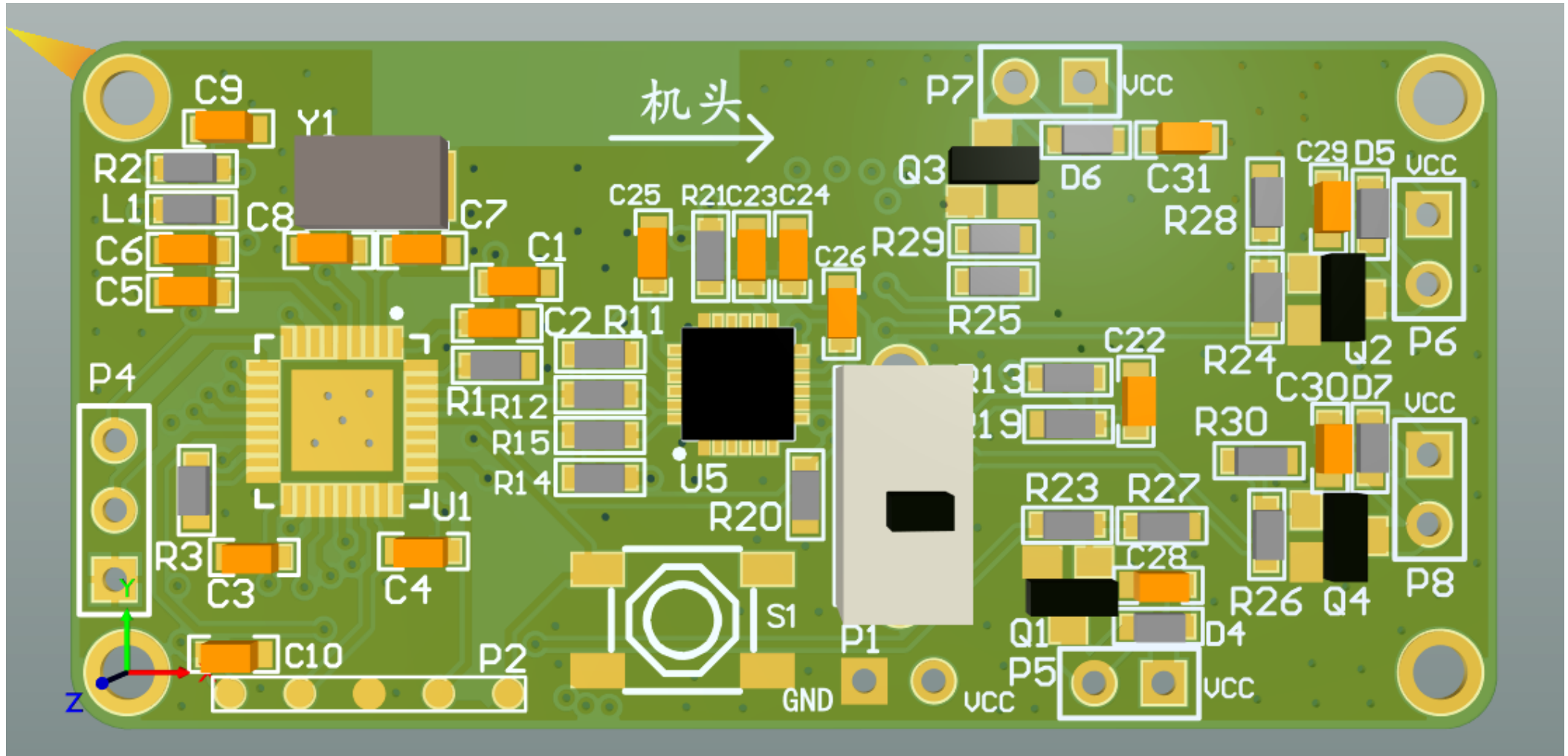
物理尺寸	25mm x 50mm (高密度微型航模规格)
PCB 叠层	4层板 (Top信号 - L2完整地平面 - L3电源走线 - Bottom底层)
供电架构	1S 航模锂电输入 → FP6276B 同步升压 5V → TP2019 高 PSRR LDO 3.3V
动力保护	4路 CJ2310 NMOS + 1N5819W 肖特基续流二极管 (LTspice 仿真钳位至 4.45V)
传感器与射频	MPU6050 居中布局, GC2400-TC017 2.4G 板载无线射频模组

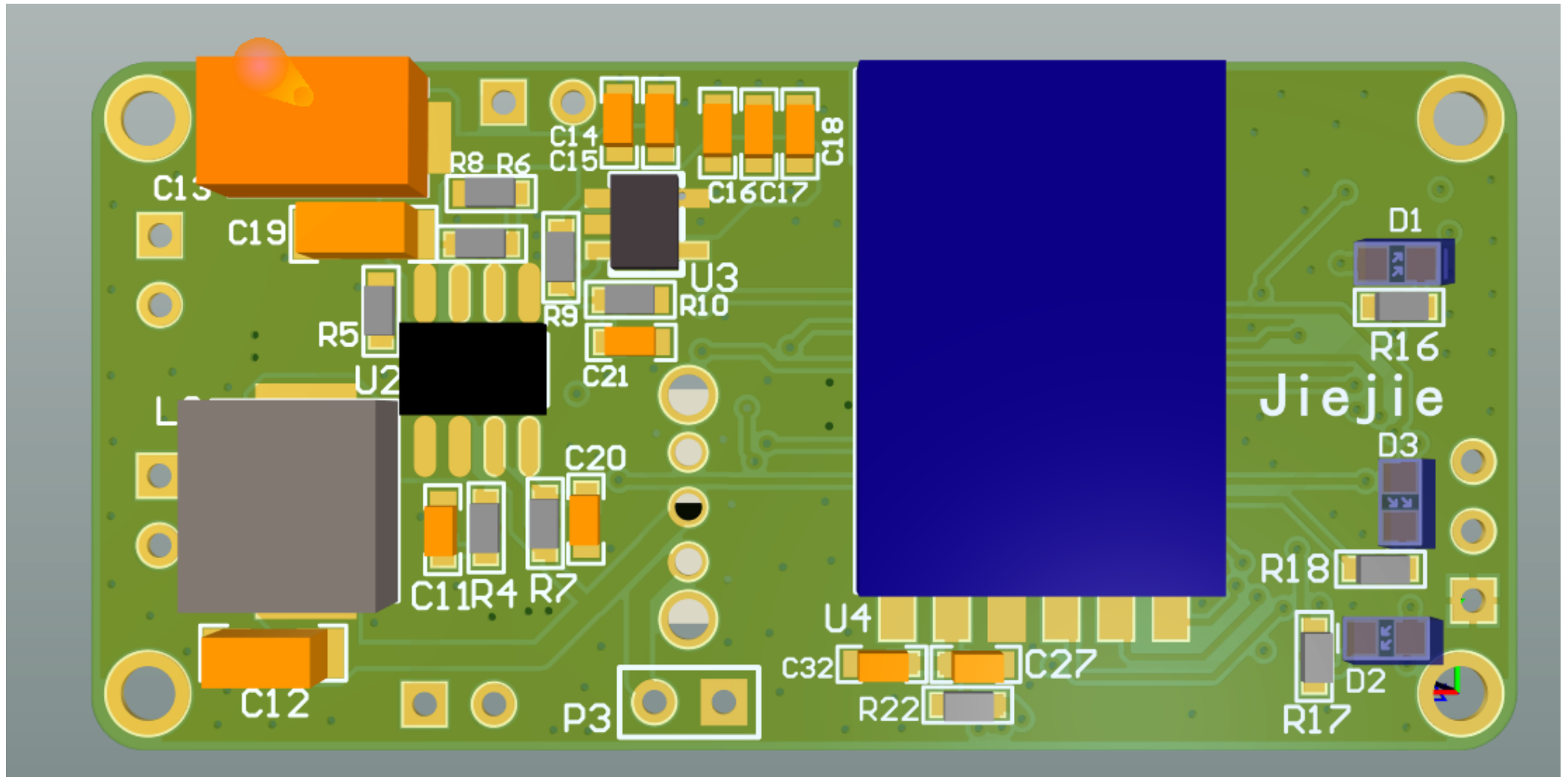
关键工程要点 / ENGINEERING HIGHLIGHTS

- 完整地回流**
L2 规划为无割裂完整地平面, 为电机 PWM 动力回路提供低阻抗返回路径。
- 强弱电隔离**
强电电机驱动阵列置于板卡右侧, 远离左侧弱电 MCU 与 IMU。
- 瞬态钳位仿真**
建立 716 空心杯电机等效模型, 将约 500V 关断感应尖峰钳位至 4.45V。

信号链摘要 / SIGNAL CHAIN

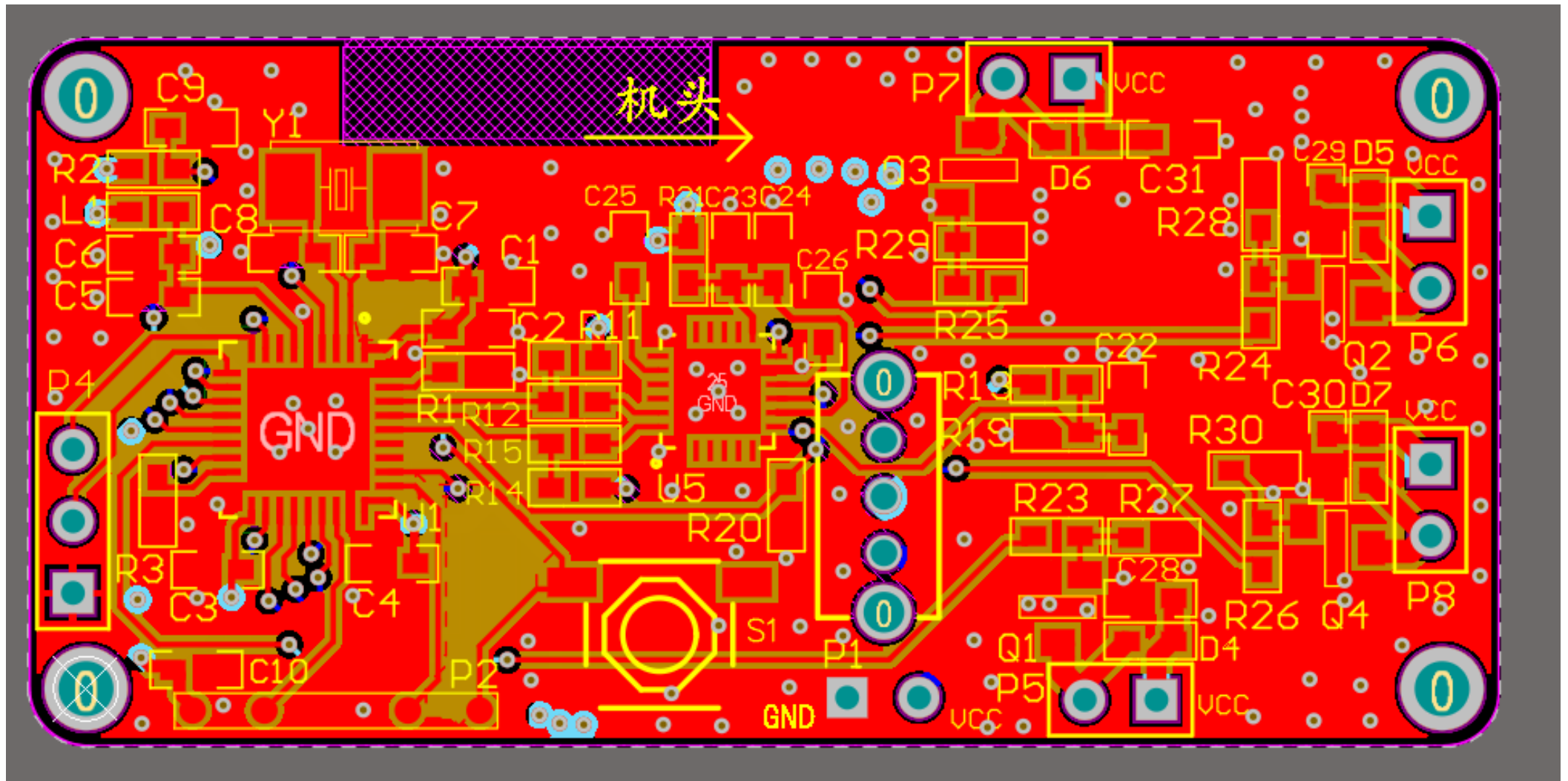


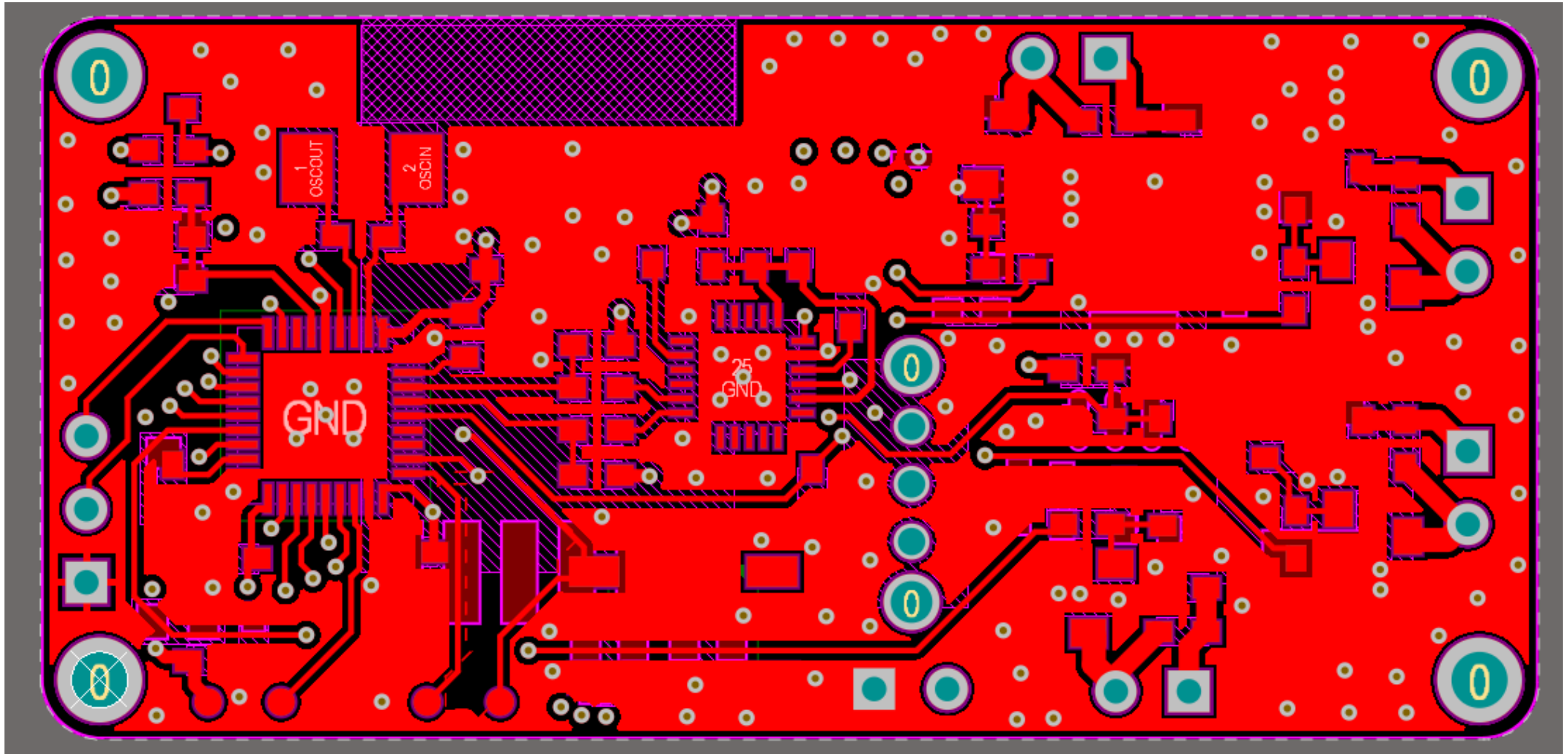


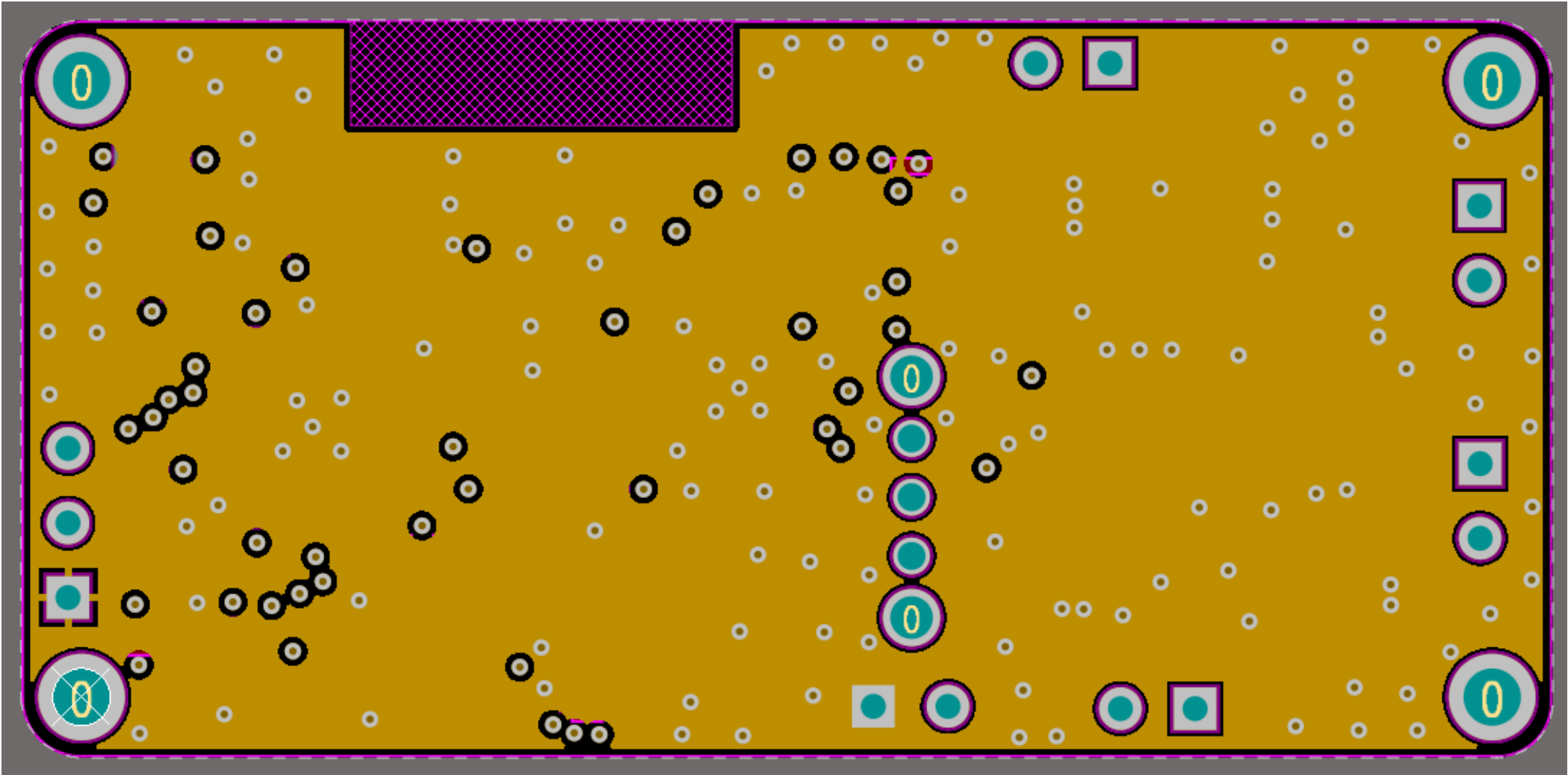


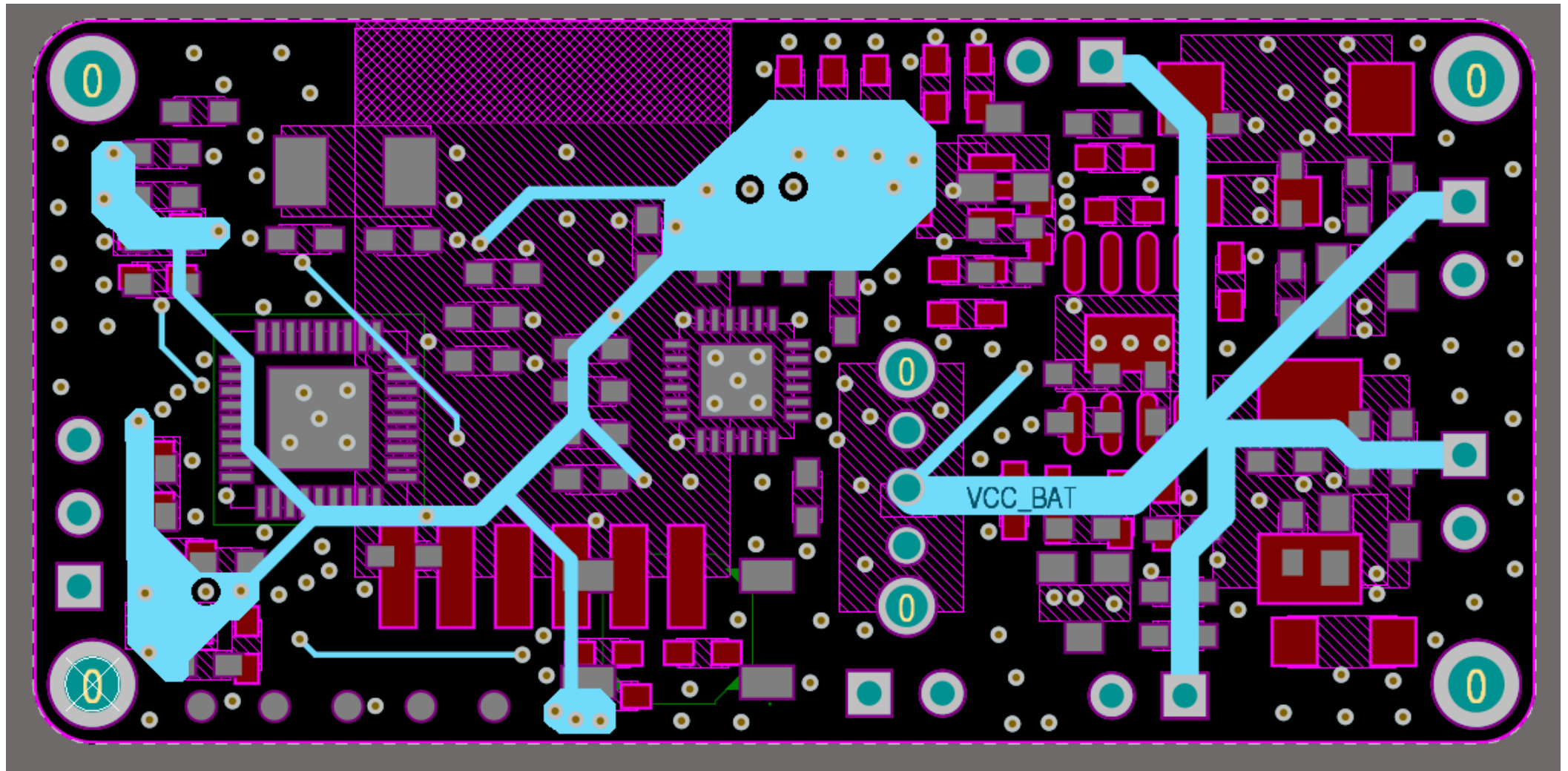


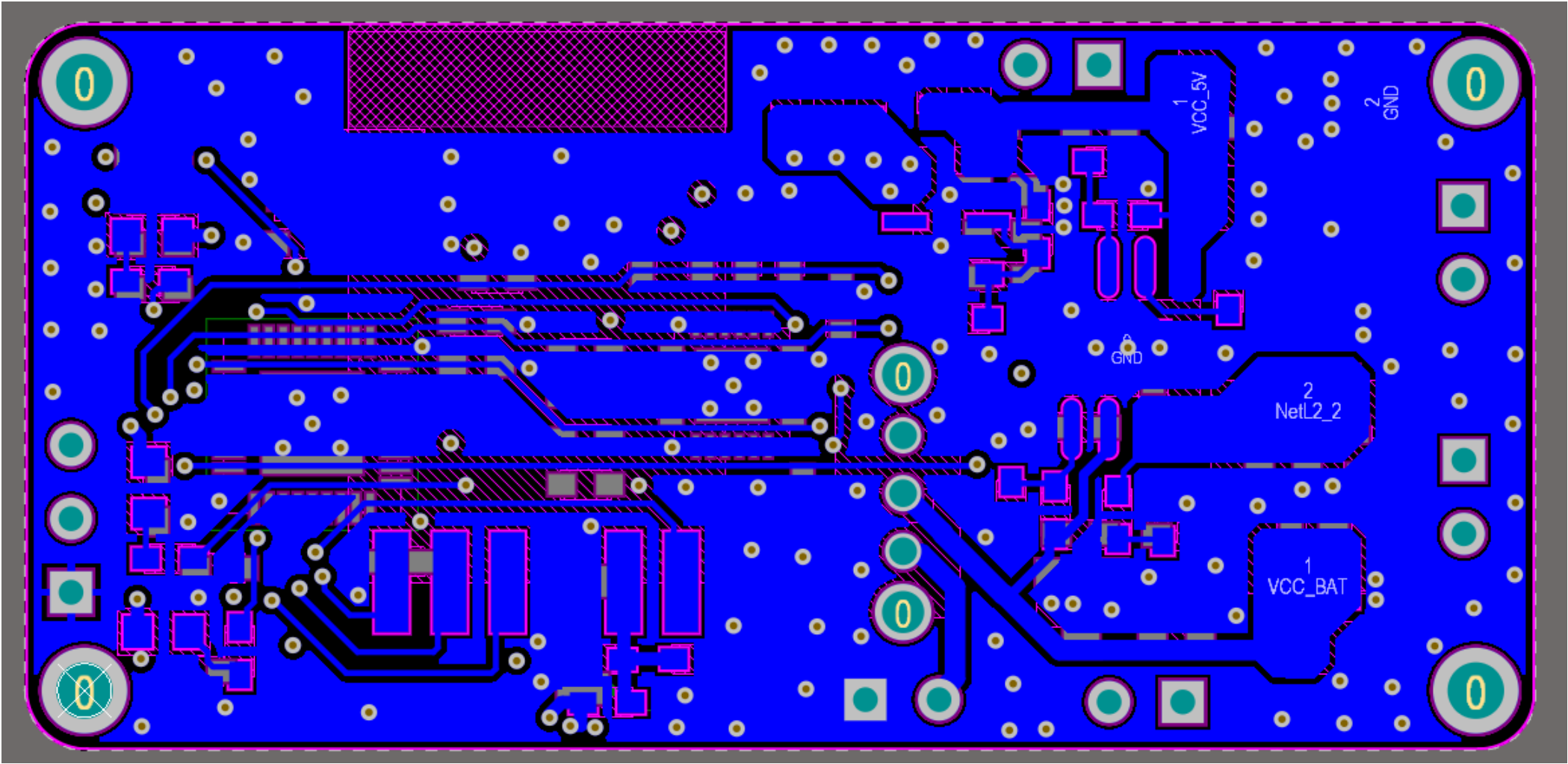


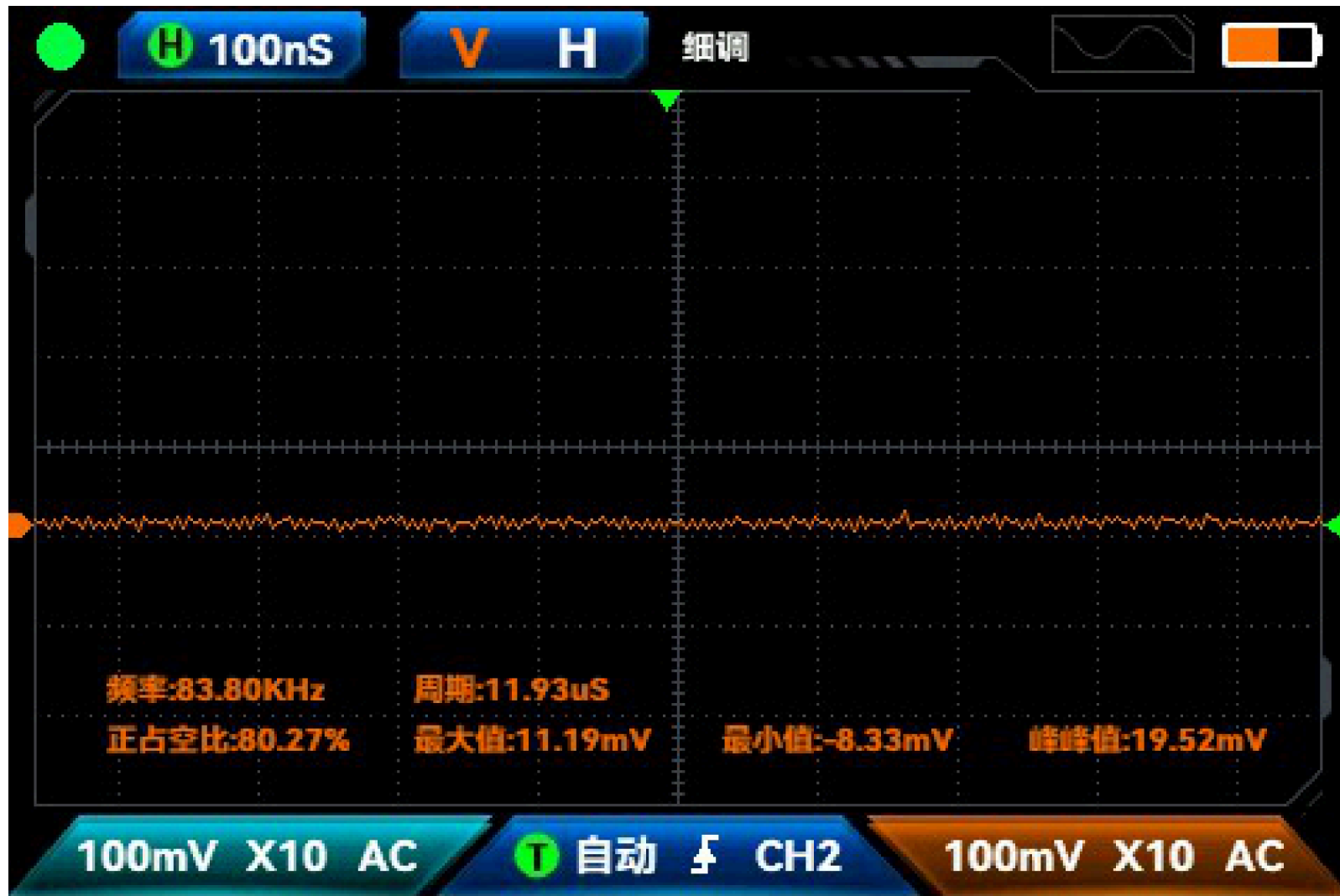




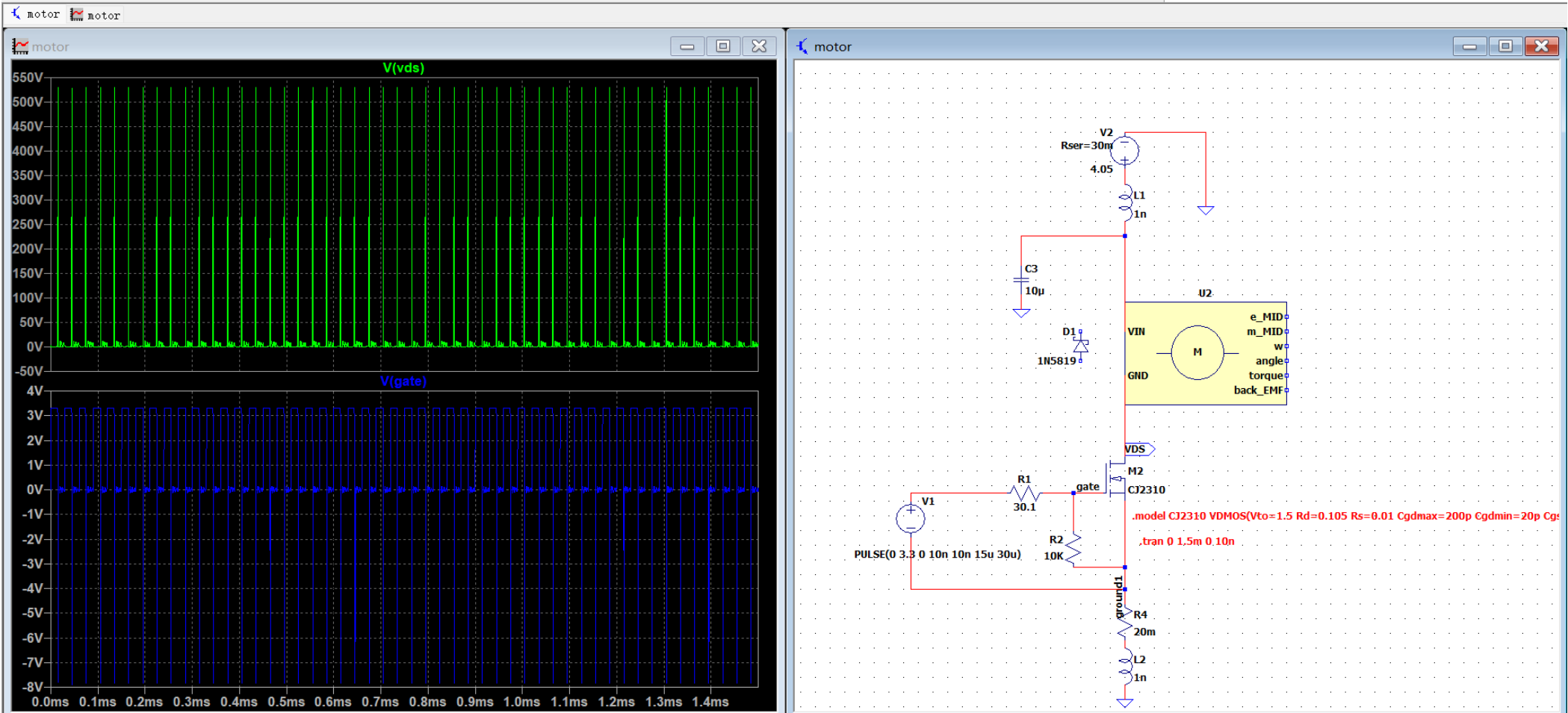






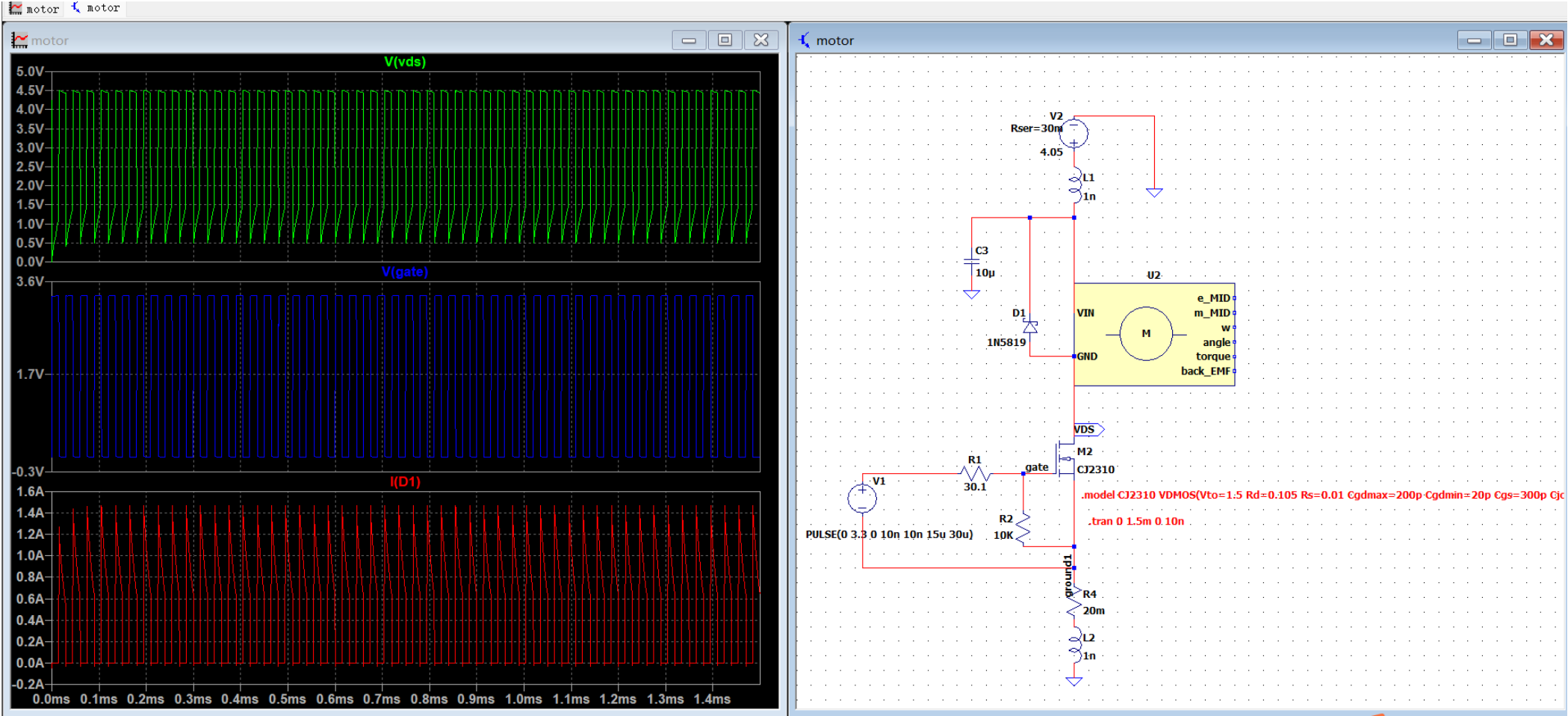


使用FNIRSI DS215H进行板级电源纹波测试，当前测试条件下测得3.3V节点Vpp约19.52mV。



CONCLUSION ·结论对比 - LTspice 瞬态仿真结果

条件 / CONDITION	峰值电压 / VDS PEAK	说明 / NOTE
Without Flyback Diode	VDS peak $\approx$ 500 V	MOSFET 关断瞬间产生高幅值感应尖峰
With 1N5819W	VDS peak $\approx$ 4.45 V	续流路径建立，关断尖峰得到有效钳位



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